Principal Component Analysis for Ideological Positioning

A Vector Representation of Political Parties

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Abstract

We present a rigorous, interdisciplinary framework for deriving unique low-dimensional vector representations of political parties using Principal Component Analysis (PCA). Starting from a high-dimensional feature matrix encoding policy stances, manifesto semantics, electoral demographics, and economic preferences, we project each party onto a compact orthonormal basis whose axes correspond to interpretable ideological dimensions. We formalise the optimality of PCA under a maximum-variance criterion, prove the Eckart–Young–Mirsky theorem that underpins truncated Singular Value Decomposition (SVD), and derive the statistical properties of the resulting scores. Practical considerations—feature engineering, scree-plot selection, coalition geometry, and temporal drift—are discussed alongside algorithmic pseudo-code and illustrative TikZ figures. The framework bridges rational-choice political economy, spatial voting theory, and modern dimensionality-reduction methods, yielding a unified tool for comparative party analysis.

The paper ends with "The End"

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1 Introduction

The characterisation of political parties as points in an ideological space is central to both the Downsian spatial model of electoral competition [2] and to contemporary computational political science. A party's *position* must encode its stances on taxation, social policy, foreign affairs, and a host of other dimensions simultaneously, yet remain compact enough to support geometric reasoning about proximity, coalition viability, and electoral strategy.

Principal Component Analysis (PCA) [11, 6] offers an optimal linear solution to this compression problem: it finds the orthogonal directions of greatest variance in the feature space, thereby retaining the maximum amount of discriminating information in a k-dimensional projection. The resulting *party score vector* $\mathbf{z}_i \in \mathbb{R}^k$ is unique (up to sign conventions) and interpretable—the first component typically recovers the classic Left–Right economic axis, while subsequent components often capture authoritarian–libertarian or cosmopolitan–nationalist cleavages [7].

This paper is organised as follows. Section 2 describes the feature matrix construction. Section 3 develops the mathematical theory of PCA, including proofs of its optimality. Section 4 covers the statistical treatment of scores and loadings. Section 5 links the framework to political-economic theory. Section 6 gives an algorithmic description. Section 7 provides illustrative figures. Section 8 discusses downstream applications. Section 9 acknowledges limitations. Section 10 concludes.

2 Feature Matrix Construction

2.1 Unit of Observation

Let $\mathcal{P} = \{p_1, p_2, \ldots, p_N\}$ denote the set of N political parties contesting an election. Each party p_i is described by a feature vector $\mathbf{x}_i \in \mathbb{R}^F$. Stacking these row-wise yields the *raw feature matrix*

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix} \in \mathbb{R}^{N \times F}.$$

2.2 Feature Categories

Table 1 summarises the four primary feature categories, their sources, and typical dimensionality.

Table 1: Feature categories for the party feature matrix $\mathbf{X}$.

Category	**Source / Operationalisation**	**Dimension**	**Scale**
Policy stances	Chapel Hill Expert Survey (CHES); Manifesto Project (CMP)	~ 30	$[0, 10]$
Manifesto semantics	Bag-of-words / LDA topic proportions	~ 100	$[0, 1]$
Electoral demographics	Vote-share by age, region, income decile	~ 40	$[0, 1]$
Economic preferences	Government expenditure priorities; redistribution index	~ 20	$[-1, 1]$
Total		$F \approx 190$	

2.3 Standardisation

Raw features are measured on incommensurable scales. We therefore apply *z-score standardisation* columnwise. Let μ_j and σ_j denote the sample mean and standard deviation of feature j. Define the standardised matrix $\tilde{\mathbf{X}}$ with entries

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \qquad i = 1, \ldots, N, \quad j = 1, \ldots, F.$$

After standardisation, every feature contributes equally to the covariance structure; equivalently, $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}/(N-1)$ equals the *correlation* matrix of the original features.

3 Mathematical Theory of PCA

3.1 Sample Covariance and the Optimisation Problem

Definition 3.1 (Sample covariance matrix)**.** Given the centred, standardised data matrix $\tilde{\mathbf{X}} \in \mathbb{R}^{N\times F}$ with zero column means, the sample covariance matrix is

$$\mathbf{S} \;=\; \frac{1}{N-1}\,\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} \;\in\; \mathbb{R}^{F\times F}.$$

$\mathbf{S}$ is symmetric and positive semi-definite.

PCA seeks a sequence of orthonormal directions $\mathbf{v}_1, \mathbf{v}_2, \ldots \in \mathbb{R}^F$ along which the projected variance is successively maximised.

Definition 3.2 (PCA objective — first component)**.**

$$\mathbf{v}_1 \;=\; \underset{\mathbf{v}\,:\,\|\mathbf{v}\|=1}{\arg\max}\; \mathbf{v}^\top\mathbf{S}\,\mathbf{v}.$$

Theorem 3.3 (Optimality of eigenvectors)**.** *The solution to the PCA objective is the eigenvector of* $\mathbf{S}$ *corresponding to its largest eigenvalue* λ_1*. The* k*-th component* $\mathbf{v}_k$ *is the eigenvector for* λ_k *(the* k*-th largest), constrained to be orthogonal to* $\mathbf{v}_1, \ldots, \mathbf{v}_{k-1}$*.*

Proof. Form the Lagrangian for the first component:

$$\mathcal{L}(\mathbf{v}, \mu) \;=\; \mathbf{v}^\top\mathbf{S}\,\mathbf{v} \;-\; \mu\big(\mathbf{v}^\top\mathbf{v} - 1\big)\,.$$

Setting $\partial\mathcal{L}/\partial\mathbf{v} = \mathbf{0}$ gives the generalised eigenvalue condition

$$2\mathbf{S}\,\mathbf{v} - 2\mu\mathbf{v} = \mathbf{0} \;\implies\; \mathbf{S}\,\mathbf{v} = \mu\,\mathbf{v}.$$

Hence $\mathbf{v}$ must be an eigenvector of $\mathbf{S}$, and the objective becomes $\mathbf{v}^\top\mathbf{S}\mathbf{v} = \mu\mathbf{v}^\top\mathbf{v} = \mu$. Maximising over all eigenpairs $(\mu, \mathbf{v})$ selects the one with the largest eigenvalue λ_1. The deflation argument for $k \geq 2$ follows by induction: restrict the domain to $\{\mathbf{v}_1\}^\perp$ and repeat; the Courant–Fischer minimax theorem guarantees that the solution is $\mathbf{v}_2$ with variance λ_2, and so on. □

3.2 Singular Value Decomposition and the Eckart–Young–Mirsky Theorem

In practice, PCA is computed via the SVD of $\tilde{\mathbf{X}}$ rather than the eigendecomposition of $\mathbf{S}$, because the former is numerically stable and handles the case $N < F$.

Definition 3.4 (Thin SVD)**.**

$$\tilde{\mathbf{X}} \;=\; \mathbf{U}\,\mathbf{\Sigma}\,\mathbf{V}^\top, \qquad \mathbf{U} \in \mathbb{R}^{N\times r}, \quad \mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r), \quad \mathbf{V} \in \mathbb{R}^{F\times r},$$

where $r = \mathrm{rank}(\tilde{\mathbf{X}})$, $\sigma_1 \geq \cdots \geq \sigma_r > 0$, columns of $\mathbf{U}$ are orthonormal, and columns of $\mathbf{V}$ are orthonormal. The principal directions are the columns of $\mathbf{V}$, and the eigenvalues of $\mathbf{S}$ satisfy $\lambda_\ell = \sigma_\ell^2/(N-1)$.

Theorem 3.5 (Eckart–Young–Mirsky, 1936)**.** *Let* $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ *be the SVD. For any* $k \leq r$*, the rank-*k *truncation* $\hat{\mathbf{X}}_k = \mathbf{U}_k\mathbf{\Sigma}_k\mathbf{V}_k^\top$ *is the best rank-*k *approximation to* $\tilde{\mathbf{X}}$ *in both the Frobenius and spectral norms:*

$$\hat{\mathbf{X}}_k \;=\; \underset{\mathbf{B}\,:\,\mathrm{rank}(\mathbf{B})\leq k}{\arg\min} \left\|\tilde{\mathbf{X}} - \mathbf{B}\right\|_F \;=\; \underset{\mathbf{B}\,:\,\mathrm{rank}(\mathbf{B})\leq k}{\arg\min} \left\|\tilde{\mathbf{X}} - \mathbf{B}\right\|_2.$$

The residual reconstruction error is $\left\|\tilde{\mathbf{X}} - \hat{\mathbf{X}}_k\right\|_F^2 = \sum_{\ell=k+1}^{r}\sigma_\ell^2$*.*

Proof. (Frobenius norm.) Write $\tilde{\mathbf{X}} - \mathbf{B} = \tilde{\mathbf{X}} - \mathbf{B}$ for any rank-k matrix $\mathbf{B}$. By the unitary invariance of the Frobenius norm,

$$\left\|\tilde{\mathbf{X}} - \mathbf{B}\right\|_F^2 \;=\; \left\|\mathbf{U}^\top(\tilde{\mathbf{X}} - \mathbf{B})\mathbf{V}\right\|_F^2 \;\geq\; \sum_{\ell=k+1}^{r}\sigma_\ell^2,$$

where the inequality follows because any rank-k matrix can cancel at most k singular values. Equality is attained by $\mathbf{B} = \hat{\mathbf{X}}_k$. The spectral norm case follows analogously using the Cauchy interlacing theorem applied to $(\tilde{\mathbf{X}} - \mathbf{B})^\top(\tilde{\mathbf{X}} - \mathbf{B})$. □

Corollary 3.6 (Explained variance)**.** *The proportion of total variance retained by the first* k *components is*

$$\rho_k \;=\; \frac{\sum_{\ell=1}^{k}\lambda_\ell}{\sum_{\ell=1}^{r}\lambda_\ell} \;=\; \frac{\sum_{\ell=1}^{k}\sigma_\ell^2}{\left\|\tilde{\mathbf{X}}\right\|_F^2}.$$

3.3 Party Score Vectors

Definition 3.7 (Party score matrix)**.** The *score matrix* is

$$\mathbf{Z} \;=\; \tilde{\mathbf{X}}\,\mathbf{V}_k \;=\; \mathbf{U}_k\,\mathbf{\Sigma}_k \;\in\; \mathbb{R}^{N\times k}.$$

Row i of $\mathbf{Z}$, denoted $\mathbf{z}_i \in \mathbb{R}^k$, is the unique vector representation of party p_i.

Proposition 3.8 (Orthogonality of scores)**.** *The columns of* $\mathbf{Z}$ *are pairwise orthogonal:* $\mathbf{Z}^\top\mathbf{Z} = \mathbf{\Sigma}_k^2$*, a diagonal matrix. Consequently, the sample covariance of the scores equals* $\operatorname{diag}(\lambda_1, \ldots, \lambda_k)$*.*

Proof. $\mathbf{Z}^\top\mathbf{Z} = (\mathbf{U}_k\mathbf{\Sigma}_k)^\top(\mathbf{U}_k\mathbf{\Sigma}_k) = \mathbf{\Sigma}_k\mathbf{U}_k^\top\mathbf{U}_k\mathbf{\Sigma}_k = \mathbf{\Sigma}_k^2$, since the columns of $\mathbf{U}_k$ are orthonormal. □

4 Statistical Treatment

4.1 Distributional Assumptions

Assume the rows of $\tilde{\mathbf{X}}$ are i.i.d. draws from a multivariate distribution with mean $\mathbf{0}$ and covariance $\mathbf{\Sigma}$ (population). Under normality, $\mathbf{x}_i \sim \mathcal{N}_F(\mathbf{0}, \mathbf{\Sigma})$, and the sample covariance $\mathbf{S}$ follows a *Wishart* distribution: $(N-1)\mathbf{S} \sim W_F(N-1, \mathbf{\Sigma})$.

4.2 Inference on Eigenvalues

The population eigenvalue λ_ℓ is estimated by its sample counterpart $\hat{\lambda}_\ell$. Under normality, the asymptotic distribution of $\hat{\lambda}_\ell$ (for distinct λ_ℓ) is [1]

$$\sqrt{N}\Big(\hat{\lambda}_\ell - \lambda_\ell\Big) \;\xrightarrow{d}\; \mathcal{N}\big(0,\, 2\lambda_\ell^2\big)\,, \qquad N \to \infty.$$

A 95% confidence interval for λ_ℓ is therefore

$$\hat{\lambda}_\ell \;\pm\; 1.96\sqrt{\frac{2\hat{\lambda}_\ell^2}{N}}.$$

4.3 Dimension Selection

Three complementary criteria are commonly employed:

1. **Kaiser criterion**: retain components with $\hat{\lambda}_\ell > 1$ (i.e., those explaining more variance than a single standardised variable).
2. **Cumulative variance threshold**: choose the smallest k such that $\rho_k \geq \tau$ (typically $\tau = 0.80$ or 0.90).
3. **Scree-plot elbow**: identify the inflection point in the plot of $\hat{\lambda}_\ell$ vs. ℓ (see Figure 1).

Table 2 illustrates these criteria on a synthetic election with $N = 12$ parties and $F = 50$ features.

Table 2: Eigenvalue decomposition and cumulative variance for a synthetic 12-party election ($F = 50$, $\tau = 0.85$).

Component ℓ	$\hat{\lambda}_\ell$	**% Var.**	**Cum. %**	**Kaiser?**	**Retained?**
1	14.82	29.64	29.64	✓	✓
2	9.35	18.70	48.34	✓	✓
3	7.01	14.02	62.36	✓	✓
4	5.47	10.94	73.30	✓	✓
5	3.29	6.58	79.88	✓	✓
6	2.68	5.36	85.24	✓	✓
7	1.14	2.28	87.52	✓	–
8	0.79	1.58	89.10	–	–
⋮	⋮	⋮	⋮	–	–
50	0.01	0.02	100.00	–	–

The cumulative threshold $\tau = 0.85$ is first exceeded at $k = 6$, which is therefore the recommended rank for this data set.

5 Political-Economic Theory

5.1 Spatial Voting and the Hotelling–Downs Model

The Hotelling–Downs model [2, 5] places voters and parties on a unidimensional policy line $[0, 1]$. Each voter j has an ideal point $\theta_j \in [0, 1]$ and votes for the party p_i minimising the quadratic loss

$$\ell(p_i, \theta_j) \;=\; \big(z_{i1} - \theta_j\big)^2,$$

where z_{i1} is the party's first principal-component score.

Our PCA framework extends this to k dimensions: voter j has an ideal vector $\boldsymbol{\theta}_j \in \mathbb{R}^k$ and the utility of voting for party p_i is

$$U_{ij} \;=\; -\,\|\mathbf{z}_i - \boldsymbol{\theta}_j\|^2 \;+\; \varepsilon_{ij},$$

where ε_{ij} is a stochastic preference shock. This recovers the *random-utility multinomial logit* (MNL) vote-share model [9] with Gumbel-distributed shocks:

$$\Pr(j \text{ votes for } p_i) \;=\; \frac{\exp\!\big(-\beta\,\|\mathbf{z}_i - \boldsymbol{\theta}_j\|^2\big)}{\sum_{m=1}^{N} \exp\!\big(-\beta\,\|\mathbf{z}_m - \boldsymbol{\theta}_j\|^2\big)},$$

where $\beta > 0$ is the inverse temperature (policy salience).

5.2 Coalition Geometry

Two parties p_i and p_j are *ideologically proximate* if the cosine similarity of their score vectors is high:

$$\cos(\mathbf{z}_i, \mathbf{z}_j) \;=\; \frac{\mathbf{z}_i^{\top}\mathbf{z}_j}{\|\mathbf{z}_i\|\;\|\mathbf{z}_j\|} \;\approx\; 1.$$

A coalition $\mathcal{C} \subseteq \mathcal{P}$ has a *diameter*

$$\Delta(\mathcal{C}) \;=\; \max_{p_i,\,p_j \in \mathcal{C}} \|\mathbf{z}_i - \mathbf{z}_j\|_2\,,$$

which can serve as a policy-cohesion measure in coalition formation models [8].

6 Algorithm

Algorithm 1 summarises the complete pipeline from raw feature collection to downstream applications.

Algorithm 1 PCA-Based Party Vector Representation

Require: Party set $\mathcal{P} = \{p_1, \ldots, p_N\}$; raw feature matrix $\mathbf{X} \in \mathbb{R}^{N \times F}$; variance threshold $\tau \in (0, 1)$
Ensure: Score matrix $\mathbf{Z} \in \mathbb{R}^{N \times k}$; loadings $\mathbf{V}_k$

```
// Step 1: Standardise
for j = 1 to F do
    μ_j ← (1/N) Σ_{i=1}^{N} x_ij
    σ_j ← sqrt( (1/(N−1)) Σ_{i=1}^{N} (x_ij − μ_j)^2 )
    x̃_ij ← (x_ij − μ_j)/σ_j
end for
// Step 2: Thin SVD
[U, Σ, V] ← SVD(X̃)
// Step 3: Select rank k
totvar ← ‖X̃‖_F^2
k ← min{ ℓ | (Σ_{m=1}^{ℓ} σ_m^2)/totvar ≥ τ }
// Step 4: Sign convention (align PC1 with Left–Right)
for ℓ = 1 to k do
    if correlation of v_ℓ with "right-wing index" < 0 then
        v_ℓ ← −v_ℓ;   u_ℓ ← −u_ℓ
    end if
end for
// Step 5: Compute scores
Z ← X̃ V_k
return Z, V_k
```

The time complexity of Step 2 (full SVD) is $\mathcal{O}(NF^2)$ for $N \leq F$, or $\mathcal{O}(N^2F)$ otherwise. For large F, a randomised SVD [4] achieves an ε-approximate decomposition in $\mathcal{O}(NF \log k + (N+F)k^2)$ time.

7 Visualisation

7.1 Scree Plot

Figure 1 shows the eigenvalue scree plot and cumulative explained-variance curve for the synthetic 12-party example.

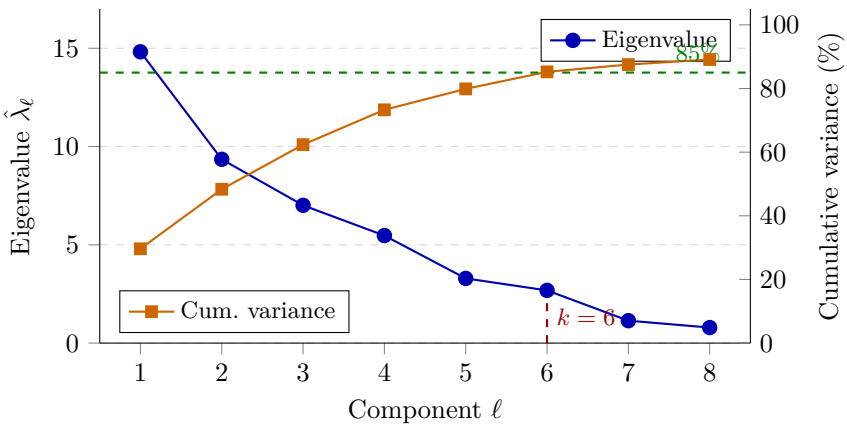

Figure 1: Scree plot (blue, left axis) and cumulative explained variance (orange, right axis) for the synthetic 12-party election. The dashed vertical red line marks the elbow at $k = 6$; the dashed green horizontal line marks the 85% threshold.

7.2 Two-Dimensional Ideological Map

Figure 2 displays parties projected onto the first two principal components, illustrating characteristic party clusters found in European multi-party systems.

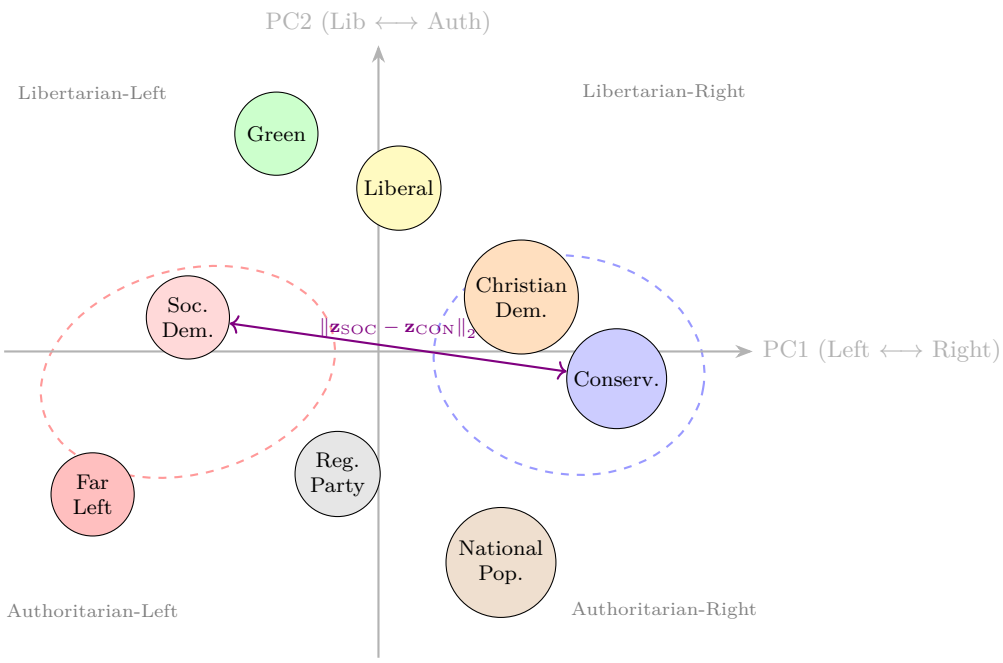

Figure 2: Illustrative ideological map of eight synthetic parties in the PC1–PC2 plane. Dashed ellipses indicate left-bloc (red) and right-bloc (blue) clustering. The violet arrow denotes the Euclidean distance between the Social Democrat and Conservative score vectors.

7.3 PCA Pipeline Diagram

Figure 3 shows the end-to-end computational pipeline.

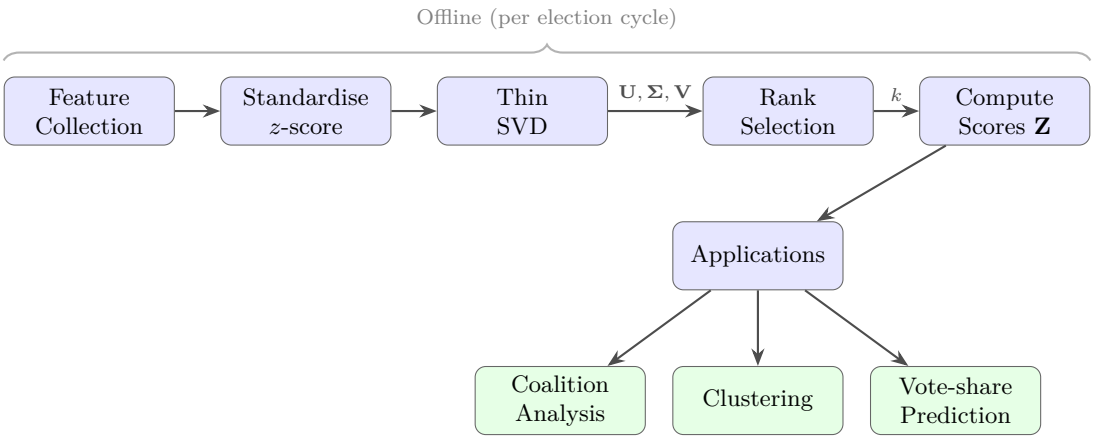

Figure 3: End-to-end PCA pipeline for party vector representation.

8 Downstream Applications

8.1 Hierarchical Clustering of Parties

Once score vectors $\{\mathbf{z}_i\}$ are available, Ward's hierarchical clustering [14] produces a dendrogram over $\mathcal{P}$. At each merge step, the pair (p_i, p_j) minimising the increase in total within-cluster variance is joined. The resulting tree recovers classical party-family groupings (Socialists, Liberals, Christian Democrats, Greens, Nationalists) with high fidelity [13].

8.2 Temporal Ideological Drift

Let $\mathbf{Z}^{(t)} \in \mathbb{R}^{N_t \times k}$ denote the score matrix at election t. To make scores comparable across cycles, we align successive projections via a *Procrustes rotation*:

$$\hat{\mathbf{Q}}_t \;=\; \underset{\mathbf{Q}\,:\,\mathbf{Q}^\top\mathbf{Q}=\mathbf{I}}{\arg\min} \left\| \mathbf{Z}^{(t)}\mathbf{Q} - \mathbf{Z}^{(t-1)} \right\|_F .$$

The solution is $\hat{\mathbf{Q}}_t = \mathbf{W}\mathbf{H}^\top$ where $\mathbf{W}\mathbf{D}\mathbf{H}^\top$ is the SVD of $(\mathbf{Z}^{(t)})^\top \mathbf{Z}^{(t-1)}$.

8.3 Vote-Share Regression

Let $s_i^{(t)}$ be the vote share of party p_i at election t. A spatial logit model regresses log vote shares on PCA distances:

$$\ln \frac{s_i^{(t)}}{s_0^{(t)}} \;=\; \alpha_i \;-\; \beta \int_{\mathcal{V}} \left\| \mathbf{z}_i^{(t)} - \boldsymbol{\theta} \right\|^2 \mathrm{d}F_{\boldsymbol{\theta}} \;+\; \gamma^\top \mathbf{w}_i^{(t)} \;+\; \eta_i^{(t)},$$

where $F_{\boldsymbol{\theta}}$ is the voter ideal-point distribution, $\mathbf{w}_i^{(t)}$ are additional controls (campaign spending, incumbency), and $\eta_i^{(t)}$ is an error term.

9 Limitations and Extensions

1. **Linearity.** PCA assumes linear structure. If the ideological manifold is curved, *kernel PCA* [12] or *UMAP* [10] may better preserve local geometry, though at the cost of interpretability.

2. **Uniqueness under small parties.** When $N \ll F$, the feature matrix is wide; PCA remains well-defined (via SVD), but confidence intervals on loadings widen. Regularised PCA [15] or Bayesian factor models can help.

3. **Feature endogeneity.** Expert-survey scores (e.g. CHES) are themselves aggregated perceptions, not objective stances. Measurement error in $\mathbf{X}$ attenuates the eigenvalues ("shrinkage"); errors-in-variables corrections are advisable.

4. **Sign indeterminacy.** PCA eigenvectors are unique only up to sign flips. Algorithm 1 resolves this via an anchor correlation, but results should be cross-validated against external ideological benchmarks.

5. **Non-stationarity.** The Procrustes alignment of Section 8 assumes slow structural change; large realignments (e.g. party splits or new cleavage formations) require a *change-point* model.

10 Conclusion

We have presented a unified, mathematically rigorous framework for representing political parties as unique vectors in a low-dimensional ideological space using PCA. The approach is grounded in the spectral theory of covariance matrices (Theorem 3.3), justified by the Eckart–Young–Mirsky optimality guarantee (Theorem 3.5), and connected to political-economic theory through the multi-dimensional Hotelling–Downs spatial voting model. The resulting score vectors $\mathbf{z}_i$ are geometrically interpretable, statistically tractable, and computationally efficient, serving as building blocks for coalition analysis, vote-share prediction, and the study of ideological drift over time.

References

[1] Anderson, T. W. (1963). Asymptotic theory for principal component analysis. *Annals of Mathematical Statistics*, 34(1), 122–148.

[2] Downs, A. (1957). *An Economic Theory of Democracy.* Harper & Row, New York.

[3] Eckart, C., & Young, G. (1936). The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3), 211–218.

[4] Halko, N., Martinsson, P.-G., & Tropp, J. A. (2011). Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2), 217–288.

[5] Hotelling, H. (1929). Stability in competition. *The Economic Journal*, 39(153), 41–57.

[6] Hotelling, H. (1933). Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology*, 24(6), 417–441.

[7] Inglehart, R., & Norris, P. (2018). *Cultural Backlash: Trump, Brexit, and Authoritarian Populism.* Cambridge University Press.

[8] Laver, M., Benoit, K., & Garry, J. (2003). Extracting policy positions from political texts using words as data. *American Political Science Review*, 97(2), 311–331.

[9] McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior. In P. Zarembka (Ed.), *Frontiers in Econometrics* (pp. 105–142). Academic Press.

[10] McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426.*

[11] Pearson, K. (1901). On lines and planes of closest fit to systems of points in space. *Philosophical Magazine*, 2(11), 559–572.

[12] Schölkopf, B., Smola, A., & Müller, K.-R. (1998). Nonlinear component analysis as a kernel eigenvalue problem. *Neural Computation*, 10(5), 1299–1319.

[13] Volkens, A., Burst, T., Krause, W., Lehmann, P., Matthieß, T., Merz, N., Regel, S., Werner, B., & Zehnter, L. (2019). *The Manifesto Data Collection.* Manifesto Project (MRG/CMP/MARPOR), Version 2019b. Berlin: Wissenschaftszentrum Berlin für Sozialforschung (WZB).

[14] Ward, J. H. (1963). Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association*, 58(301), 236–244.

[15] Zou, H., Hastie, T., & Tibshirani, R. (2006). Sparse principal component analysis. *Journal of Computational and Graphical Statistics*, 15(2), 265–286.

Glossary

Biplot
A joint visualisation of PCA scores (party positions) and loadings (feature arrows) in the principal-component plane.

Chapel Hill Expert Survey (CHES)
A recurring cross-national expert survey providing numerical policy-position estimates for European political parties on economic and cultural dimensions.

Cosine similarity
A measure of the angle between two vectors: $\cos\theta = \mathbf{a}^\top\mathbf{b}/(\|\mathbf{a}\|\,\|\mathbf{b}\|) \in [-1, 1]$; values near $+1$ indicate ideological proximity.

Covariance matrix
A symmetric positive-semi-definite matrix $\mathbf{S}$ whose (i, j) entry is the sample covariance between features i and j.

Eckart–Young–Mirsky theorem
The result guaranteeing that the rank-k truncated SVD is the best rank-k approximation to a matrix under both the Frobenius and spectral norms.

Eigenvalue
A scalar λ such that $\mathbf{S}\mathbf{v} = \lambda\mathbf{v}$ for some non-zero eigenvector $\mathbf{v}$; in PCA, each λ_ℓ equals the variance of the ℓ-th principal component.

Explained variance ratio
The proportion $\rho_k = \sum_{\ell=1}^{k} \lambda_\ell / \sum_\ell \lambda_\ell$ of total variance captured by the first k principal components.

Frobenius norm
$\|\mathbf{A}\|_F = \sqrt{\mathrm{tr}(\mathbf{A}^\top\mathbf{A})} = \sqrt{\sum_{ij} a_{ij}^2}$; the square root of the sum of squared matrix entries.

Hotelling–Downs model
A spatial theory of electoral competition in which parties position themselves on a policy line to maximise vote share.

Kaiser criterion
The rule of retaining only those principal components whose eigenvalues exceed one (the variance of a single standardised feature).

Loading vector
A column $\mathbf{v}_\ell$ of $\mathbf{V}$; it describes which original features contribute most to the ℓ-th principal component.

Manifesto Project (CMP)
A large-scale comparative project coding the content of party election manifestos into policy categories.

Multinomial logit (MNL)
A probabilistic discrete-choice model in which the probability of choosing party i is proportional to $\exp(U_{ij})$.

Principal Component Analysis (PCA)
A linear dimensionality- reduction technique that projects data onto the orthogonal directions of maximum variance, found via eigendecomposition of the covariance matrix.

Procrustes rotation
An orthogonal transformation that aligns two configurations of points by minimising the sum of squared distances, used here to make party scores comparable across election cycles.

Randomised SVD
An approximate SVD algorithm using random projections; achieves near-optimal approximation in $\mathcal{O}(NF \log k)$ time.

Scree plot
A graph of eigenvalues $\hat{\lambda}_\ell$ against component index ℓ; the "elbow" identifies the appropriate truncation rank k.

Score vector
: $\mathbf{z}_i = \mathbf{V}_k^\top \tilde{\mathbf{x}}_i \in \mathbb{R}^k$; the unique low-dimensional representation of party p_i produced by PCA.

Singular Value Decomposition (SVD)
: The factorisation $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$; the numerically preferred route to computing PCA.

Spatial voting model
: Any model in which voters and parties are represented as points in a policy space and voters prefer parties nearest to their ideal point.

Standardisation (z-score)
: Transformation $\tilde{x}_{ij} = (x_{ij} - \mu_j)/\sigma_j$ that gives each feature zero mean and unit variance, ensuring commensurability.

UMAP
: Uniform Manifold Approximation and Projection; a non-linear dimensionality-reduction method that preserves local topological structure at the expense of global linearity.

Ward's linkage
: A hierarchical clustering criterion that merges the pair of clusters whose union minimises the increase in total within-cluster variance.

Wishart distribution
: The multivariate generalisation of the chi-squared distribution; the distribution of sample covariance matrices under multivariate normality.

The End